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object_tracking_node:
  ros__parameters:
    # --- Node settings ---
    debug: false
    state: 1 # NodeState.ACTIVE
    export_timing_csv: false # CSV timing export on shutdown (default:
off)
    crossing_tracking_enabled: false # set to false to disable the
crossing tracker entirely

    # --- Subscriber topics ---
    image_subscriber: "/camera/image/undistorted"
    object_detection_subscriber: "/object_detection/object"
    sign_detection_subscriber: "/object_detection/sign"
    crossing_detection_subscriber: "/crossing_detection/result"
    state_subscriber: "/tracking/state"

    # --- Publisher topics ---
    state_publisher: "/tracking/state"

    # --- Common parameters ---
    dt: 0.1 # Fallback time step (s), only used for
first frame
    publish_interval_ms: 15 # Publisher-frequency (ms)
    tracker_step_interval_ms: 40 # Min. interval (ms) for timer-based
predict step when no detection arrives
    max_x: 3000.0 # Max valid x position (mm) - forward
range
    max_y: 2000.0 # Max valid y position (mm) - lateral
range

    # --- Object tracker (moving objects: cars, pedestrians) ---
    object_max_age: 32 # Max frames without update before
deletion
    object_min_hits: 6 # Min successful updates for
confirmation
    object_min_age: 5 # Min frames for confirmation
    object_max_distance: 4.5 # Max Mahalanobis distance (3.03:  $\chi^2$ 
99%, 2 DoF)
    object_q_pos: 50.0 # Process noise: position (mm)
    object_q_vel: 180.0 # Process noise: velocity (mm/s)
    object_r_pos: 100.0 # Measurement noise: position (mm) at
reference distance
    object_r_dist_ref: 1400.0 # Reference distance for R scaling (mm)
- higher = less distance effect, lower = more
    object_sigma_pos_init: 100.0 # Initial position uncertainty (mm)
    object_sigma_vel_init: 500.0 # Initial velocity uncertainty (mm/s)
    object_duplicate_distance: 0.0 # Min Euclidean distance (mm) below
which duplicate tracks are suppressed (0 = disabled)

    # --- Sign tracker (traffic signs) ---
    sign_max_age: 35 # Max frames without update before

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deletion
    sign_min_hits: 3                # Min successful updates for
confirmation
    sign_min_age: 2                 # Min frames for confirmation
    sign_max_distance: 7           # Max Mahalanobis distance (3.03:  $\chi^2$ 
99%, 2 DoF)
    sign_q_pos: 50.0               # Process noise: position (mm)
    sign_q_vel: 250.0              # Process noise: velocity (mm/s) –
covers apparent velocity changes due to ego-motion
    sign_r_pos: 150.0              # Measurement noise: position (mm) at
reference distance
    sign_r_dist_ref: 1400.0        # Reference distance for R scaling (mm)
– higher = less distance effect, lower = more
    sign_sigma_pos_init: 100.0     # Initial position uncertainty (mm)
    sign_sigma_vel_init: 500.0     # Initial velocity uncertainty (mm/s) –
high because ego-velocity at track creation is unknown
    sign_duplicate_distance: 0.0    # Min Euclidean distance (mm) below
which duplicate tracks are suppressed (0 = disabled)

# --- Crossing tracker (lane lines at intersections) ---
    crossing_max_age: 3            # Max frames without update before
deletion
    crossing_min_hits: 1           # Min successful updates for
confirmation
    crossing_min_age: 0            # Min frames for confirmation (0 =
confirm as soon as min_hits is reached)
    crossing_max_distance: 4.5     # Max Mahalanobis distance (3.03:
 $\chi^2$  99%, 2 DoF)
    crossing_q_pos: 20.0           # Process noise: position (mm)
    crossing_q_vel: 30.0           # Process noise: velocity (mm/s)
    crossing_r_pos: 80.0           # Measurement noise: position (mm) at
reference distance
    crossing_r_dist_ref: 1100.0    # Reference distance for R scaling
(mm) – higher = less distance effect, lower = more
    crossing_sigma_pos_init: 300.0 # Initial position uncertainty (mm)
    crossing_sigma_vel_init: 300.0 # Initial velocity uncertainty (mm/s)
    crossing_duplicate_distance: 0.0 # Min Euclidean distance (mm) below
which duplicate tracks are suppressed (0 = disabled)

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